Assignment – 1.4

Day 4: GPUs and Cooling Systems Theory (30 min):Basics of GPUs and cooling systems (air vs. liquid cooling).Practical (30 min):Identify a GPU and cooling system in a PC setup.

Task – 1

Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

Objective

The objective of this task is to identify the Graphics Processing Unit (GPU) installed in my computer and determine its brand, model, and type.

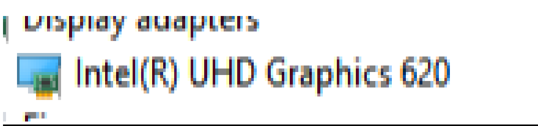
GPU Identification

I checked my computer using **Windows Device Manager**. Under the **Display adapters** section, the installed graphics processor is shown as **Intel(R) UHD Graphics 620**.

GPU INFORMATION	DETAILS
GPU Brand	Intel
GPU Model	Intel(R) UHD Graphics 620
GPU Type	Integrated GPU
Identification Method	Device Manager

Screenshot

Figure 1: Identifying the installed GPU using Device Manager



About Intel UHD Graphics 620

The **Intel UHD Graphics 620** is an integrated graphics processor commonly found in Intel laptop processors. It is designed for everyday graphics tasks such as displaying the Windows desktop, watching videos, browsing the internet, and running lightweight applications.

Since it is an **integrated GPU**, it does not operate as a separate graphics card with its own dedicated VRAM. Instead, it can use a portion of the computer's system memory when required.

It is suitable for normal computer usage and some light gaming, but it is not designed to provide the same level of performance as modern dedicated GPUs used for high-end gaming, 3D rendering, and professional graphics applications.

Task – 2

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

➔

1. Objective

The objective of this task is to identify the main components of a computer cooling system and understand how they help maintain a safe operating temperature inside a PC.

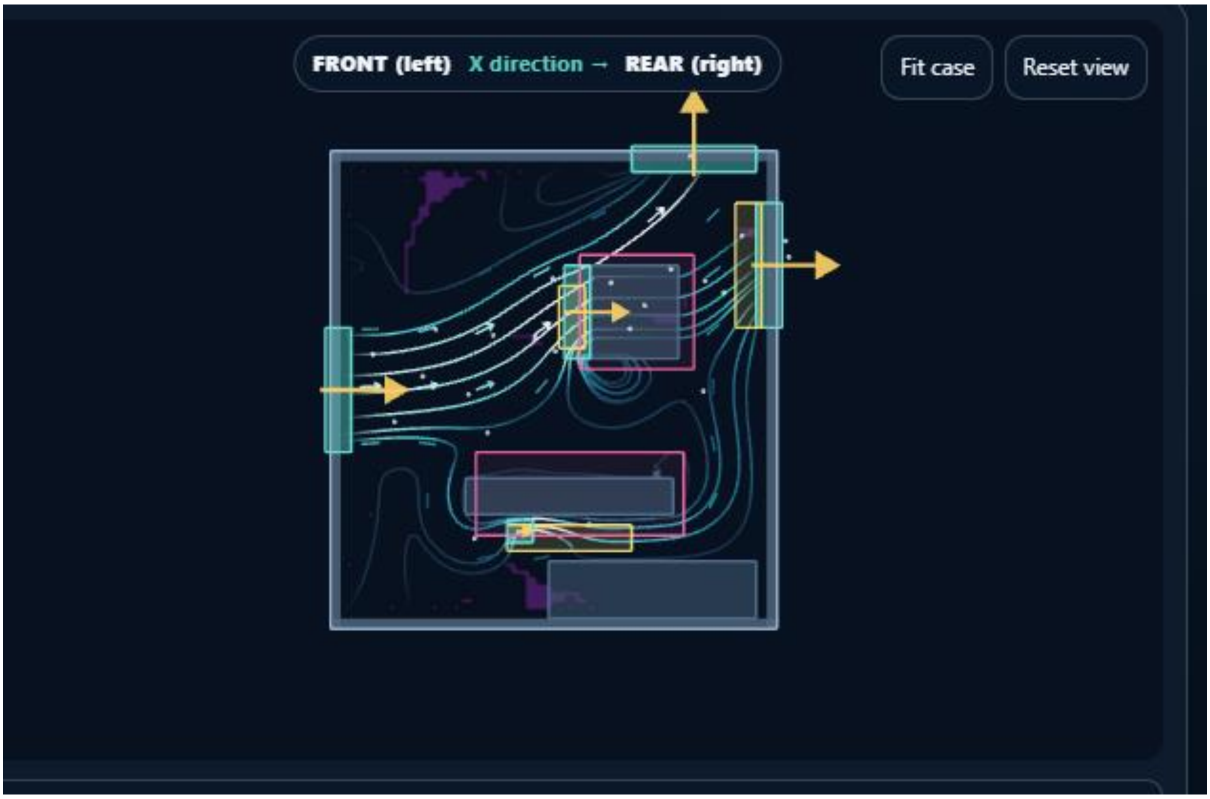
2. Cooling System Used

For this task, I used a **PC hardware airflow simulator** to observe the cooling system and airflow inside a computer case. I selected a **front intake and rear exhaust** configuration.

In this arrangement, air enters the computer through the front fan and moves through the internal components. The heated air is then pushed out through the rear and top exhaust areas. This continuous airflow helps remove heat from the computer.

3. Cooling System Screenshot

Figure 1: PC cooling system and airflow using a hardware simulator



Main Cooling Components

Component	Function
Cooling Fan	Moves air through the PC case to bring in cooler air and remove warm air.
Air Vent	Provides an opening for air to enter or leave the computer case.
Heatsink	Absorbs and spreads heat away from components such as the CPU or GPU.
Heatpipe	Transfers heat from a hot component to a heatsink in many cooling designs.
Airflow	Carries heat away from internal components and maintain proper temperature

How Cooling Works

The cooling process starts when the **intake fan** brings relatively cool air into the computer case. The air passes over and around components that generate heat, such as the CPU and

GPU. The heated air then travels toward the **exhaust fan or vent**, where it is removed from the case.

Good airflow is important because excessive heat can reduce component performance and may shorten the life of computer hardware. Therefore, fans, heatsinks, vents, and proper airflow work together to keep the system operating efficiently.

Task – 3

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

➔

Objective

The objective of this task is to compare **air cooling and liquid cooling systems** used for GPUs based on cost, cooling efficiency, performance, and maintenance.

2. What is Air Cooling?

Air cooling uses a combination of a **heatsink and fan** to remove heat from the GPU. The heatsink absorbs heat from the GPU and spreads it across its fins. The fan then moves air across the heatsink to carry the heat away.

Air cooling is commonly used because it is **simple, affordable, and easy to maintain**.

3. What is Liquid Cooling?

Liquid cooling uses a liquid coolant to transfer heat away from the GPU. The coolant absorbs heat from the GPU through a cooling block and carries it to a **radiator**, where fans remove the heat. The cooled liquid then circulates back to the cooling block.

Liquid cooling can provide **better heat management**, especially for high-performance systems.

Comparison

Factor	Air Cooling	Liquid Cooling

Cost	Generally cheaper	Generally more expensive
Cooling Efficiency	Good for normal workloads	Usually better for high heat loads
Performance	Suitable for most GPUs	Suitable for high-performance GPUs
Installation	Simple and easy	More complex
Maintenance	Low	Requires more attention
Noise	Noisy under heavy load	Can be quitier

Task-4

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research

➔

For a gaming PC built to stream IPL matches and handle graphics-heavy games, liquid cooling is the better choice for your GPU. It provides superior thermal management under sustained high loads, keeping temperatures lower and preventing throttling during long sessions. Air cooling is simpler and reliable, but liquid cooling ensures quieter operation and higher performance consistency when dealing with powerful GPUs like the RTX 40-series. This makes liquid cooling ideal for smooth gameplay and uninterrupted streaming.